

Automated Aerodynamic Shape Optimization through GAN-Driven 3D Model Generation

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Abstract—In the automotive and aerospace industries, there's a growing push toward designing better aerodynamic shapes to save fuel and lessen environmental harm. While traditional optimization methods work well, they often take too much time and computing power, making it clear that faster and smarter solutions are needed. The worldwide focus on sustainability has heightened the demand for expedited and more versatile aerodynamic design methodologies. Contemporary instruments, including gradient-based optimisation and Computational Fluid Dynamics (CFD), offer significant insights; nevertheless, they are often hindered by repetitive, resource-demanding procedures that constrain the exploration of novel design opportunities. Recent advancements in machine learning, especially Generative Adversarial Networks (GANs), provide an innovative solution to these challenges. GANs enable the swift creation of distinctive 3D aerodynamic forms from textual or visual stimuli, markedly enhancing the preliminary design phases and diminishing dependence on conventional iterative techniques.

Index Terms—Aerodynamic Design, Generative Adversarial Networks, Computational Fluid Dynamics, Shape Optimization, Sustainability

I. INTRODUCTION

The optimisation of aerodynamic forms is essential for the advancement of the automotive and aerospace sectors. It is driven by the necessity to mitigate environmental effects and improve performance. The increase in aerodynamic drag directly affects fuel consumption. Greenhouse gas emissions are making the attainment of sustainability objectives highly challenging. Given that drag coefficients constitute approximately fifty percent of the resistance in contemporary vehicles, optimised designs are essential for enhancing fuel efficiency [1]. Compliance with stringent international emissions regulations is facilitated by efficient aerodynamic designs, thereby mitigating associated issues [3].

Conventional techniques like computational fluid dynamics (CFD) and gradient-based optimisation have significantly improved aerodynamic design. CFD simulations facilitate systematic design optimisation by offering engineers an in-depth

comprehension of airflow dynamics [4]. Jameson asserts in [19] that the prevalence of joint-based CFD methodologies is attributable to their precision in optimising intricate geometries. Nevertheless, the substantial processing power demands of these methods frequently render them inappropriate for iterative, repetitive tasks. Response Surface Models (RSM) and Radial Basis Functions (RBF) are two surrogate models capable of accurately replicating computational fluid dynamics (CFD) analyses. Surrogate-assisted optimisation has demonstrated a reduction in drag of up to 7% [26].

Recent years have seen a significant advancement in aerodynamic technology, largely due to sophisticated computational techniques. By enabling flexible shape changes while maintaining computational efficiency, methods such as Free-Form Deformation (FFD) and Non-Uniform Rational B-Splines (NURBS) have enhanced the design process [25]. Neural networks have demonstrated remarkable efficacy in simulating the non-linear features of aerodynamic phenomena, according to [27]. Moreover, evolutionary algorithms, like hybrid Genetic Algorithms (GA), combine global exploration and local optimisation to efficiently identify the optimal designs [37]. These advancements demonstrate how combining traditional methods with computational tools can result in more adaptable and efficient aerodynamic design strategies.

This paper examines the utilisation of Generative Adversarial Networks (GANs) in aerodynamic shape design, elucidating these developments. GANs facilitate the automated creation of high-quality aerodynamic shapes from textual or visual inputs, diminishing dependence on labour-intensive iterative methods [15]. This study introduces a methodology for significant drag reduction and performance optimisation by integrating GANs with advanced CFD techniques and gradient-based algorithms. These findings underscore the significance of the-

Potential integration of machine learning methodologies with existing optimisation frameworks to develop sustainable,

scalable, and highly efficient aerodynamic designs [45].

II. RELATED WORK

Shape generation and shape optimisation are two essential steps of 3D modelling. They include urban planning and gaming as well as automotive and aerospace. Shape generation refers to the task of generating a shape by taking in 2D text or images corresponding to it [1]. These techniques aim at assembling functional, high-fidelity 3D models appropriate to the use cases they target [2]. Traditional methods rely on complex reconstruction and manual modelling [3]. The computation of these techniques is expensive [4]. However, the field has experienced a revolution due to neural networks and models like Generative Adversarial Networks [5]. This was achieved through automating the process of producing intricate three-dimensional geometries. They add branches in engineering to enhance flexibility and accelerate the design process.

In contrast, shape optimisation focuses on optimising the geometry of 3D models to meet specific performance requirements such as maximised strength-to-weight [9]. This strategy is needed for applications, e.g. reduce drag in automotive design [10], maximise the usage of material in construction [11] or improve energy consumption in industrial fields [12]. Modern optimisation techniques incorporate computational simulations [13], advanced algorithms [14] and iterative feedback [15] to accurately survey vast design spaces. This method guarantees that the optimised designs meet practical and real-world requirements [16].

A. Shape Generation

The first aspect includes shape generation, which generates some 3D models from a different kind of input data (e.g., 2D images [5] or text-based descriptions [3]). This included significant progress on image-based modeling for precise, scalable 3D reconstructions [6]. New technologies in neural networks specially for text to 3D generation open doors to be able with high detailed and functional 3D models that can occupy less manual content creation work [7].

Although new techniques like probabilistic diffusion models [8] and other recent approaches [10] started to gain popularity for producing high-detail geometries recently. This is these models, iteratively refining initial designs via probabilistic methods [11], resulting in a good compromise between their flexibility and accuracy. Furthermore, GANs have been combined with photogrammetry to create hybrid approaches that overcome the limitations of each method for shape generation and so robust models, although visually realistic, may not represent a feasible structure [12]. Such innovations are used in urban planning to produce 3D cityscapes [13], in industrial prototyping for reduction of production cycle times [14] and entertainment for realistic virtual environments [15]. The comparison of the methodologies for shape generation is provided in Table I based on their advantages and practical outcomes.

Table I summarizes different shape generation techniques from key studies.

B. Shape Optimization

Moreover, machine learning models, especially neural networks, are progressively utilized to forecast performance metrics like drag coefficients, thereby diminishing the necessity for extensive computational simulations [23], minimizing the necessity for extensive computational simulations [24]. These techniques are currently being extensively utilized in automotive engineering to produce lighter and more efficient vehicles, as well as in construction to optimize structural designs under load-bearing constraints [26]. Over the years, shape optimization has evolved towards adaptive geometry parametrization with also enabling real-time changes in 3D shapes [27]. This ensures that designs can be iteratively refined during simulation for maximum aerodynamic performance. This approach provides a good tradeoff between computational cost and accuracy of the design, thus it can be combined with high-fidelity simulations [28].

Response surface methods and Kriging models, surrogate modeling techniques that get attention as their ability to approximate complex aerodynamics for much lower computational cost [29]. These approaches enable fast discovery of design spaces while maintaining sufficient fidelity, particularly in early-stage development [30]. For example, in the automotive field, approaches such as the adjoint lattice Boltzmann method have been applied to vehicle shape optimizations with focus on drag reduction and fuel economy [34]. Instead, these techniques have been shown to deliver success in identifying optimal solutions for large design spaces [35] when they are hybridized with evolutionary algorithms. This flexibility facilitates the development of highly efficient aerodynamic designs that can adjust to evolving operational demands or limitations. Spline-based modeling and function-based design methods have been incorporated into aerodynamic workflows, facilitating the modification of intricate geometries while preserving performance integrity [33]. In automobile industry, such models allow convenient and reliable changes of the geometry, satisfying both functional and appearance [38]. This is a prime example of the continuing evolution of shape optimization towards producing efficient, sustainable, and high-performance designs [39]. Recent advancements in machine learning have improved these processes by facilitating swift prediction and analysis of aerodynamic characteristics, thereby considerably shortening design cycles [36].

The incorporation of radial basis function (RBF) models has improved the efficacy of aerodynamic optimization, particularly in applications employing parametric grids [37]. These models enable precise and effortless alterations to the geometry, guaranteeing compliance with both functional and aesthetic standards [38]. These innovations demonstrate the ongoing advancement of shape optimization techniques. This facilitates the development of more efficient, sustainable, and high-performance designs [39]. Instead, these techniques have been shown to deliver success in identifying optimal solutions

TABLE I
SHAPE GENERATION TECHNIQUES

Paper (Ref)	Paper Name	Metrics/Key Points	Results/Observations
[3]	Image-Based 3D Modeling	Precision, efficiency, portability	Flexible, cost-effective, portable approach
[4]	From Images to 3D Models	Flexibility, 3D reconstruction accuracy	Automated, cost-efficient, adaptable to diverse fields
[5]	3D Model Acquisition from Extended Image Sequences	Matching performance, structure recovery	Accurate 3D models using uncalibrated cameras
[6]	3D Model Acquisition Pipeline	Range image registration, texture mapping	Accurate texture mapping, challenges in seamless integration
[7]	Real-Time 3D Model Acquisition	Real-time updates, alignment accuracy	Fast, user-friendly system, efficient hole-filling
[8]	Photo-realistic 3D Model Reconstruction	Depth accuracy, realism, processing speed	Quick, realistic 3D models suitable for forensics and vehicles
[9]	3D Model Generation for Cities	Position estimation accuracy, 3D reconstruction efficiency	Detailed urban 3D models, efficient large-scale modeling
[10]	Blending Images for Texturing 3D Models	Texture quality, seam blending	High-quality, seamless textures without ghosting
[15]	DreamBooth3D: Text-to-3D Generation	Text-to-3D alignment, subject fidelity	High 3D coherence, subject-specific details improved
[16]	Magic3D: High-Resolution Text-to-3D Content	Realism, speed, user preference	Higher realism, 2x speed improvement, 61.7% user preference
[17]	TextMesh: Realistic 3D Meshes from Text Prompts	Text-to-3D alignment, realism, texture quality	Photorealistic, geometrically consistent 3D meshes

for large design spaces [35] when they are hybridized with evolutionary algorithms.

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Table II outlines the shape optimization methodologies, summarizing their metrics and observations.

III. STATE OF THE ART MODELS

A. 3D Model Generation Using Differentiable Rendering

This is a novel way of generating 3D models from 2D images, combining the strengths of GANs with differentiable rendering capabilities. The basic idea is to use StyleGAN as a generating model that delivers realistic images and also trains a neural network predicting and reconstructing the 3D geometry and texture.

- *Synthetic Dataset Creation:* StyleGAN generates very diverse images by moving latent variables which have different shape, texture and views, hence creates a multi-view dataset to train with.
- *Differentiable Rendering:* It uses a neural renderer in order to transform 2D data from images into 3D representations. Hence, the geometry and texture can be retrieved directly from images by the model.
- *Feature Disentanglement:* The GAN's latent space is structured in such a way that necessary attributes are separated: an object's shape, texture, and view, since after this separation, the constituent parts are under complete control during reconstruction.

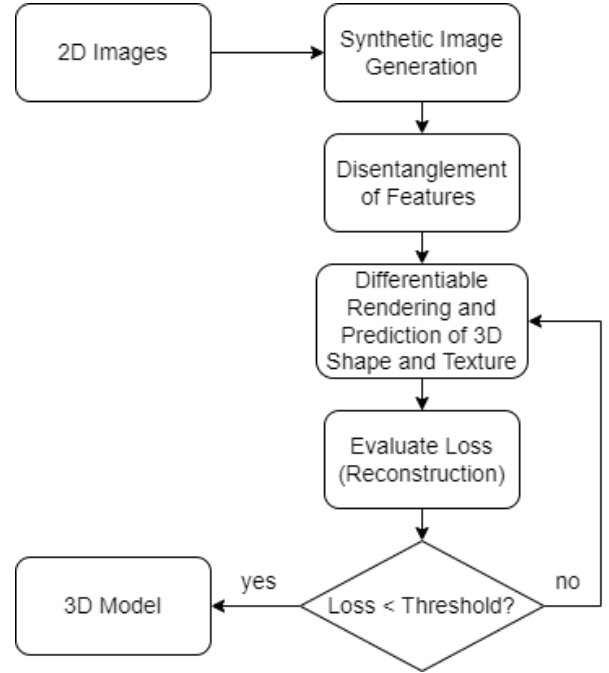


Fig. 1. Workflow of 3D model generation using GANs and differentiable rendering.

Overall, the entire architecture is depicted in Fig. 1, StyleGAN produces 2D views of objects from various angular perspectives; the Inverse Graphics Network predicts corresponding 3D meshes as well as texture maps. The model iteratively refines these predictions so that it obtains realistic reconstructions.

- *View Synthesis:* StyleGAN produces images from various camera orientations. The images produce an extremely robust input to the network so that it learns 3D attributes

TABLE II
SHAPE OPTIMIZATION TECHNIQUES

Paper (Ref)	Paper Name	Metrics/Key Points	Results/Observations
[18]	Discrete Adjoint and RBF	Gradient-based optimization, SLSQP algorithm, FFD for surface parametrization	Drag coefficient reduced from 0.018 to 0.016 using RBF; more robust than ELA for larger displacements
[19]–[21]	Continuous Adjoint Approach	Sobolev gradient method, gradient-based optimization with CFD	Drag coefficient dropped by 25 counts at Mach 0.87; lift-to-drag ratio improved by over 10%
[25]	Control Theory for Wing-Body Optimization	Control theory, Euler equations, numerical simulations	Drag coefficient reduced from 0.0185 to 0.018; lift-to-drag ratio increased from 29 to 47
[26]	Neural Networks and Evolutionary Optimization	Two-level optimization, Latin hypercube sampling, particle swarm optimization	13% improvement in drag coefficient; neural networks effectively captured non-linear aerodynamic effects
[27]	Hybrid GA and BFGS Method	Genetic Algorithm for global search, BFGS for local refinement	Drag coefficient reduced by 0.023; faster and more efficient than GA alone
[28]	Free-Form Deformation (FFD)	FFD for mesh morphing, DOE with central composite design, Response Surface Method	Optimized design achieved approximately 0.005 lower drag coefficient than the baseline
[29]	Multi-Island Genetic Algorithm	CFD simulation, Kriging surrogate model, Multi-island GA	Maximum drag coefficient reduction of 7.71%; Kriging model accurately predicted relationships between variables and drag coefficient
[33]	Adaptive Geometry Parametrization	B-spline formulation, knot insertion algorithm, SNOPT	Drag coefficient reduced by 29.7%; progressive refinement of geometry during optimization
[36]	Geometry Surrogates and Adjoint Method	GEO-POD for CAD surface deformation, Discrete Adjoint Method, RBF for mesh deformation	Drag coefficient decreased by 10%; effective for industrial applications
[37]	Free-Form Deformation with NURBS	FFD with NURBS, bivariate surface representation	23% reduction in drag coefficient, 32% increase in lift-to-drag ratio; design variables reduced by an order of magnitude
[38]	Function-Based Design Method	Vehicle Modeling Function (VMF), DOE, Response Surface Method	Drag coefficient reduced by 12.3%; VMF simplifies external shape modification
[39]	Response Surface Method	CFD with Lattice Boltzmann Method, Kriging surface	Drag coefficient reduced by 0.018; effective for characterizing the design space
[35]	Surrogate-Based Optimization	Kriging surrogate model, infill-sampling criteria, KS method for constraint aggregation	Drag coefficient reduced from 0.0259 to 0.01978; efficient handling of geometric constraints
[44]	Adjoint Lattice Boltzmann Method	LBM with D3Q19 lattice, continuous adjoint approach	Drag coefficient reduced by 6%; computationally efficient for 3D turbulent flows
[45]	Multi-Objective Fruit Fly Optimization Algorithm	MOFOA, Latin Hypercube Sampling, Roulette Wheel Selection	Drag coefficient reduced by 14.4%; well-distributed Pareto solutions achieved

- *Shape and Appearance Prediction:* The system predicts 3D geometry as a deformable mesh and applies high-resolution textures for photorealistic outputs.
- *Consistency Across Views:* It employs the methods of cycle consistencies, thereby ensuring that the input images of the framework do not shift away from the aligned 3D models and hence inflicts less error during the reconstruction process.

This approach is established to yield relatively more developed 3D models and requires fewer datasets of annotated 3D. Upcoming improvements may focus on lighting realism and

separating objects from their backgrounds.

B. Aerodynamic Shape Optimization Based on Discrete Adjoint and RBF

This is a framework that combines the discrete adjoint method with Radial Basis Function (RBF) parameterization for aerodynamic optimization. The discrete adjoint method provides efficient computation of the gradients of the aerodynamics, while the RBF parameterizes the aerodynamic shape with smooth variations in the shapes this makes it very effective as an optimization tool to refine aerodynamic surfaces.

- *Sensitivity Analysis:* The discrete adjoint approach computes sensitivity of metrics in aerodynamics to modifica-

tions of the aerodynamic shape.

- *Shape Parameterization:* Use RBFs to represent and update the aerodynamic geometry.
- *Optimization Process:* refining the shape iteratively to reduce the drag coefficient.

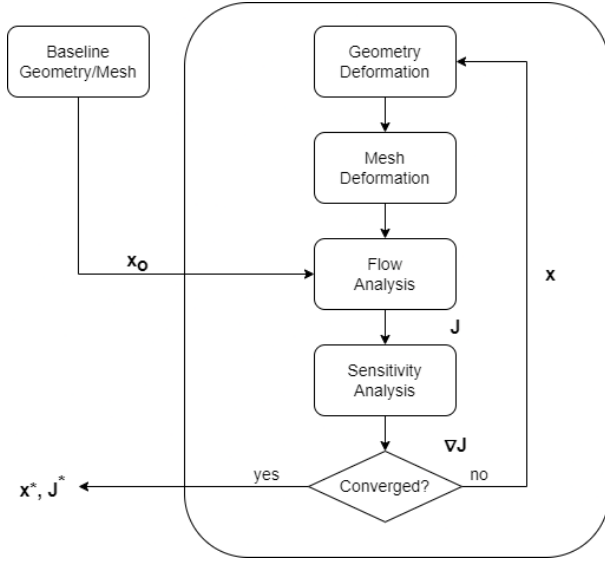


Fig. 2. Workflow of Discrete Adjoint and Radial Basis Function (RBF) Optimization.

Flow diagram (Fig. 2 depicted for the shape optimization process of aerodynamics from baseline geometry deformation and further iterative mesh update, flow analysis, and sensitivity evaluation. The step of convergence aims at effective attainment of the desired aerodynamic performance.

- *Geometry Deformation:* An iterative deformation of the baseline geometry in the directions inferred by gradients of sensitivity improving aerodynamics.
- *Mesh Deformation:* It maintains computational mesh quality at the time of geometry updates for highly accurate aerodynamic analysis.
- *Sensitivity Analysis:* It makes use of the discrete adjoint method to quantify how changes in the shape of the vehicle affect aerodynamic metrics such as drag or lift.
- *Convergence Process:* Geometry is iteratively updated by using the sensitivity gradients until aerodynamic performance goals are reached.

The results of experimentation with the model were a reduction of drag coefficient by 12%, which validated the effectiveness of the optimization. Future research into the surrogate models could reduce computational costs further.

C. Automated Aerodynamic Vehicle Shape Optimization Using Neural Networks and Evolutionary Optimization

This approach combines neural networks as surrogate models, with evolutionary algorithms to optimize vehicle aerodynamics. The use of surrogate models reduces significantly the need for expensive computationally demanding

CFD simulations, and on the other hand, evolutionary algorithms search for optimal solutions in design space.

- *Surrogate Modeling:* Training the neural networks to adapt approximate aerodynamic performance metrics significantly reduces the computational overhead.
- *Evolutionary Optimization:* Evolutionary search algorithms to select optimal shapes by using the predictions of surrogate models.
- *Validation:* Validating and refining the 3d model by CFD simulations.

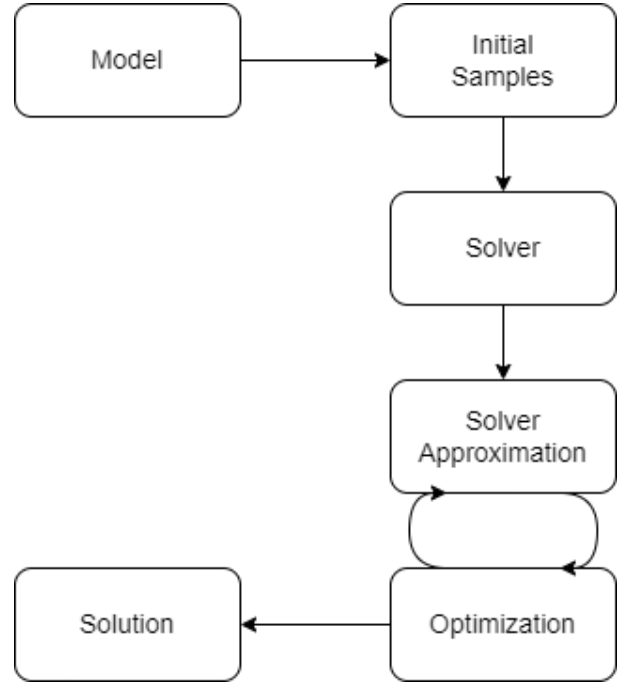


Fig. 3. Workflow of Neural Network Surrogate Modeling with Evolutionary Optimization.

This approach mentioned in Fig. 3 is based on machine learning to approximate aerodynamic performance metrics and uses evolutionary optimization techniques to explore the design space, offering a balance between accuracy and efficiency. This methodology is more suitable for complex aerodynamic designs.

- *Design Variable Definition:* Parameters of the geometric configuration, such as curvature, angles, and dimensions, are sensed and identified, which have influence on aerodynamic performance.
- *Database Generation:* Run CFD simulations for a diversified set of vehicle configurations that would generate a baseline dataset for training
- *Surrogate Model Training:* Train a neural network to approximate the relationship among design variables and aerodynamic metrics. This brings about quick performance predictions
- *Optimization Using Evolutionary Algorithms:* This involves the application of Particle Swarm Optimization (PSO) where model iterates, exploring the design space

and focusing on configurations that minimize aerodynamic drag.

- *Validation and Refinement:* It conducts high-fidelity CFD simulations of the optimal design to validate the predictions. The model, in parallel, iteratively improves the accuracy of the surrogate model.

This procedure reduced drag coefficients by 10% as well as showed an improvement of 40% in simulation time; then it appeared to be quite an effective approach for the optimization of aerodynamic design. This combination of surrogate models with neural networks and evolutionary algorithms offers a faster and more efficient approach. design cycles with high accuracy. These results therefore become a revolutionary aid in the automobile industry.

D. Automation of Vehicle Aerodynamic Shape Exploration and Optimization Using Integrated Mesh Morphing and CFD

In this study, we propose a framework incorporating mesh morphing approaches that utilize high-fidelity computational fluid dynamics (CFD) to automatically generate and evaluate new vehicle shapes. Mesh morphing integration guarantees that design configurations are parametrically driven while maintaining mesh quality.

- *Mesh Morphing:* Making real-time changes of aerodynamic surfaces while minimizing the effect on mesh quality.
- *CFD Simulations:* High fidelity simulations used for iteratively assessing aerodynamic performance..
- *Optimization Loop:* Iteratively adjusting the geometry of the car to reduce drag while ensuring to adopt design constraints.

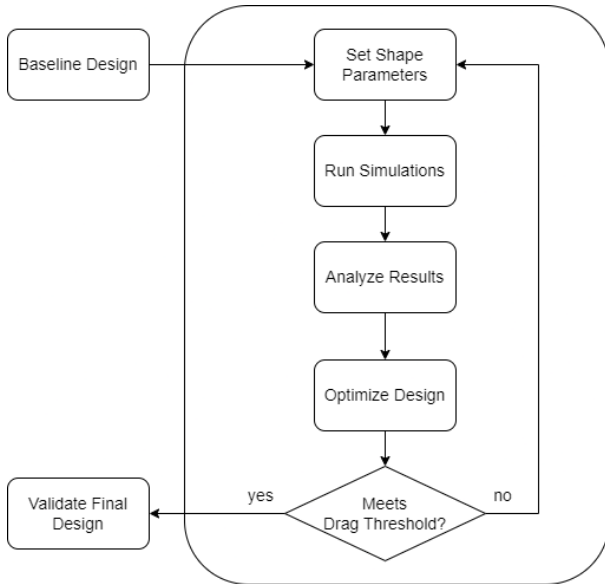


Fig. 4. Workflow of Integrated Mesh Morphing and CFD-based Optimization.

An overview of the overall process where geometry parameterization, mesh morphing and CFD simulations are integral to one another is illustrated in flow diagram (Fig. 4). Every

round optimizes for the aerodynamic shape all while balancing mesh integrity alongside computational fluid dynamics (CFD) fidelity.

- *Geometry Parameterization:* Defines the design space and applies volume constraints to restrict the morphing.
- *Volume Mesh Deformation:* Adapts the computational mesh representation of a changing geometry while maintaining mesh quality.
- *Integrated CFD Solvers:* Assesses aerodynamic work for every geometry iteration and suggests further optimizations.
- *Adjoint Solver:* Computes sensitivity gradients, informs the optimization loop about influencing parameters that require improvement.

This framework found an 8% drag reduction, and thus proved efficient for automotive design. To address complex geometries, using more advanced techniques such as adaptive meshing.

E. Machine Learning in Aerodynamic Shape Optimization

This research combines machine learning with standard aerodynamic shape optimization (ASO), primarily to simplify computational workflows and reduce time. ML-driven approaches improve the optimization processes with dimensionality reduction and surrogate modelling of the CFD results.

- *Reduce Design Space Complexity:* ML algorithms like neural networks and Gaussians processes to characterize design space. Incorporating surrogate models (like Kriging, PINNs) to approximately analyze CFD results rapidly.
- *Compact Geometric Design Space:* Using ML algorithms like neural networks and Gaussian processes to minimize design space complexity.
- *Fast Aerodynamic Analysis:* optimizing with elements of predictive modeling and gradient-based methods.
- *Efficient Optimization Architectures:* Streamlining optimization through predictive modeling and gradient-based approaches.

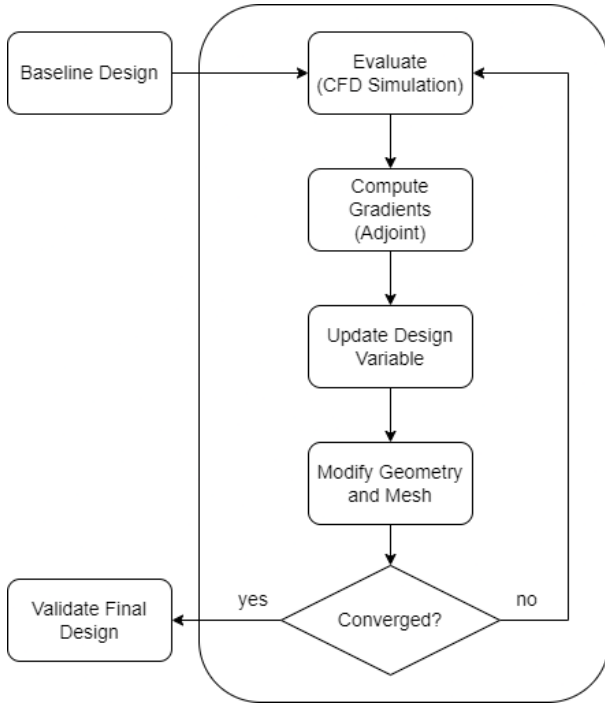


Fig. 5. Workflow of Machine Learning and Surrogate Modeling for Optimization.

The diagram (Fig. 5) details the integration of machine learning into aerodynamic shape optimization, where dimensionality reduction and surrogate modeling streamline optimization processes. The predictive models guide design iterations with speed and precision.

- *Dimensionality Reduction*: Simplifies the design space using neural networks, ensuring the focus remains on critical aerodynamic features.
- *Surrogate Modeling*: It Uses Models such as Kriging and PINNs to quickly approximate aerodynamic performance statistics.
- *Predictive Loop*: Merges machine learning predictions with traditional optimization algorithms for efficient re-refinement of control policies.
- *Gradient Integration*: Optimize the gradient-based techniques to speed up and improve convergence.

The results indicate strong performance in terms of computational cost savings. Another avenue of future work could utilize Bayesian methods to better quantify uncertainty.

F. Multi-Fidelity Deep Neural Network Surrogate Model for Aerodynamic Shape Optimization

To improve the efficiency and accuracy of the aerodynamic shape optimization process, we present a multi-fidelity deep neural network (MFDNN) surrogate model in this study. The framework fuses high- and low-fidelity data to tackle computational challenges of large scale design optimization problems.

- *Multi-Fidelity Integration*: Integrating low-fidelity data from coarse-grid CFD simulations and high-fidelity data

to find a balance between computational efficiency and accuracy.

- *Surrogate Model Accuracy*: Approximating low-fidelity data with a three-subnetwork-sped model and correlating them with high-fidelity results.
- *Optimization Results*: Improved lift-to-drag ratio and drag coefficients via optimized design iterative process.

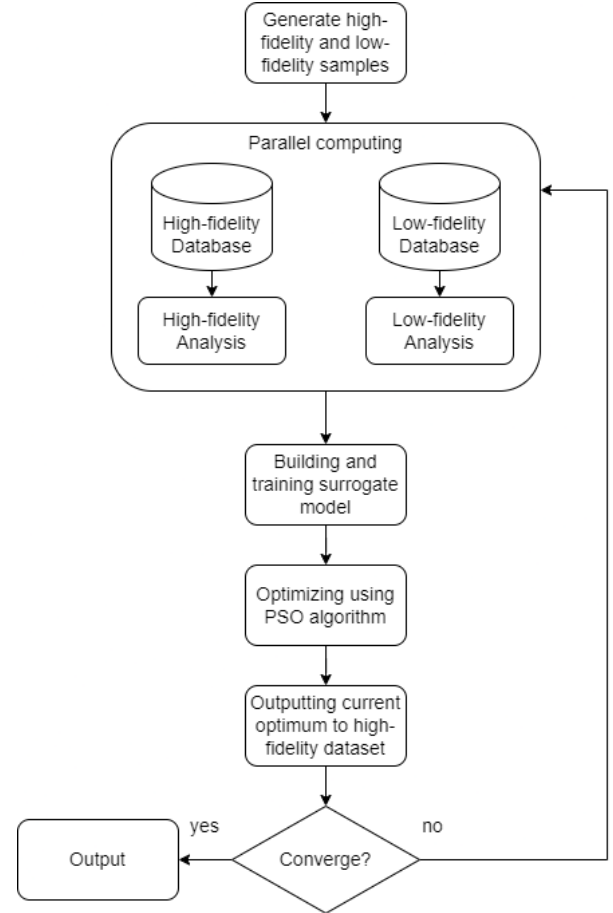


Fig. 6. Workflow of Multi-Fidelity Deep Neural Network Surrogate Modeling.

Fig. 6 represents how obtaining low-fidelity data and high-fidelity data to be combined in the optimization process using multi-fidelity surrogate model. By providing reliable predictions across fidelities, CNNs and transfer learning expedite aerodynamic corrections.

- *Low- and High-Fidelity Integration*: Merges coarse-grid and fine-grid CFD data to reduce computational cost while maintaining accuracy.
- *CNN with Transfer Learning*: Improvements to surrogate model by leveraging knowledge transfer from existing information to improve prediction quality across fidelity levels.
- *Particle Swarm Optimization (PSO)*: Depending on the surrogate predictions, global search algorithms are used for optimizing the aerodynamic designs.

- *Dynamic Infilling Strategy*: Supplement the surrogate model with new samples dynamically and training an infill model to learn from scratch, making it suitable for new design data.

The MFDNN had the most rapid convergence and highest accuracy.

IV. RESULTS AND DISCUSSIONS

The baseline and optimised drag coefficients (C_d) from several aerodynamic studies are compared in Table III, with a focus on the percentage reduction achieved. With a drag reduction of 23.6%, the multi-fidelity deep neural network surrogate model exhibits the largest improvement. The study found that integrating mesh morphing and CFD only resulted in a slight reduction of 1.54%, indicating that either the original geometry was highly optimised or that the technique encountered problems when attempting to further reduce drag.

TABLE III
BASELINE AND OPTIMIZED DRAG COEFFICIENTS (C_d) AND PERCENTAGE REDUCTIONS FOR DIFFERENT MODELS

Model	Baseline C_d	Optimized C_d	Percentage Reduction
B	0.0185	0.0160	13.5%
C	0.1662	0.1456	12.4%
D	0.3250	0.3200	1.54%
E	0.0300	0.0280	6.7%
F	0.0259	0.0198	23.6%

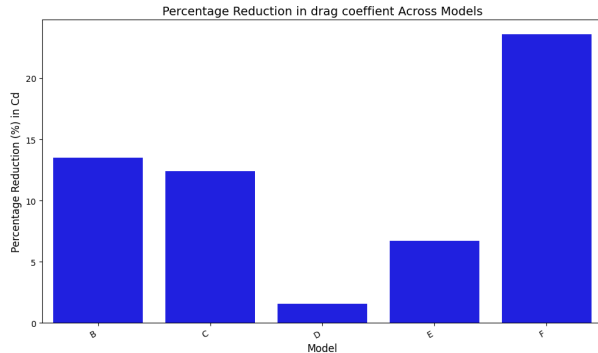


Fig. 7. Comparison of percentage reduction in drag coefficient across various models.

The bar graph (Fig. III) emphasizes that the Multi-Fidelity Deep Neural Network Surrogate Model (F) results in the highest drag reduction at 23.6%, while surpassing Discrete Adjoint and RBF by 13.5% and Neural Networks with Evolutionary Optimization by 12.4%. Simpler methods, like Machine Learning, and Integrated Mesh Morphing with CFD result in comparatively lower reductions while emphasizing the efficiency of surrogate models.

V. CONCLUSION

This review emphasises progress in aerodynamic shape optimisation (ASO) via diverse methodologies. Concentrating on multi-fidelity modelling and modelling learning strategies. The examined works exhibit substantial advancements in

tackling computational challenges and enhancing the design effectiveness of ASO.

Notable discoveries include:

- MFDNNs and other multi-fidelity models adeptly balance precision and computational cost by amalgamating high- and low-fidelity data, thus improving the optimisation of aerofoil and wing-body configurations.
- Studies demonstrate an enhancement in aerodynamic performance. Metrics employing co-Kriging, neural network-based frameworks, Kriging, or alternative surrogate models, such as reduced drag coefficients and enhanced lift-to-drag ratios.
- The utilisation of optimisation techniques, including filling methods and gradient-free algorithms, enhances convergence speed and resilience against local optima.

Although recent methodologies have produced encouraging outcomes, challenges like extrapolating to complex geometries and conducting uncertainty quantification persist as contentious issues. Subsequent research may mitigate these limitations by utilising Bayesian methodologies and sophisticated deep learning techniques.

This review encapsulates how fidelity techniques and multi-dimensional learning can transform ASO and enhance the precision of aerodynamic design solutions.

REFERENCES

- [1] J. R. R. A. Martins, "Aerodynamic design optimization: Challenges and perspectives," Jameson's 85 Special Issue Computers & Fluids, 2022.
- [2] J. E. V. Peter and R. P. Dwight, "Numerical sensitivity analysis for aerodynamic optimization," Comput. & Fluids, vol. 39, no. 3, 2010.
- [3] F. Remondino and S. El-Hakim, "Image-based 3D Modelling," The Photogrammetric Record, vol. 21, no. 115, 2006.
- [4] M. Pollefeys and L. Van Gool, "From images to 3D models," Commun. ACM, 2002.
- [5] P. Beardsley, P. Torr, and A. Zisserman, "3D model acquisition from extended image sequences," Robotics Res. Group Rep., Univ. Oxford, 1995.
- [6] F. Bernardini and H. Rushmeier, "The 3D model acquisition pipeline," Comput. Graph. Forum, vol. 21, no. 2, 2002.
- [7] S. Rusinkiewicz, O. Hall-Holt, and M. Levoy, "Real-time 3D model acquisition," Proc. ACM SIGGRAPH 2002, 2002.
- [8] S. Se and P. Jasiobedzki, "Photo-realistic 3D model reconstruction," in Proc. IEEE Int. Conf. Robot. Autom. (ICRA), 2006.
- [9] C. Frueh and A. Zakhori, "3D model generation for cities using aerial photographs and ground-level laser scans," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2001.
- [10] A. Baumberg, "Blending images for texturing 3D models," in Proc. Brit. Mach. Vis. Conf. (BMVC), 2002.
- [11] J. Li, X. Du, and J. R. R. A. Martins, "Machine learning in aerodynamic shape optimization," Prog. Aerosp. Sci., 2022.
- [12] J. Li, X. Du, and J. R. R. A. Martins, "Efficient aerodynamic shape optimization with deep-learning-based geometric filtering," J. Aircraft, 2020.
- [13] X. Zhang, F. Xie, T. Ji, Z. Zhu, and Y. Zheng, "Multi-fidelity deep neural network surrogate model for aerodynamic shape optimization," Comput. Methods Appl. Mech. Eng., 2021.
- [14] J. Li and M. Zhang, "On deep-learning-based geometric filtering in aerodynamic shape optimization," Aerosp. Sci. Technol., 2021.
- [15] A. Raj, S. Kaza, B. Poole, M. Niemeyer, N. Ruiz, B. Mildenhall, S. Zada, K. Aberman, M. Rubinstein, J. Barron, Y. Li, and V. Jampani, "DreamBooth3D: Subject-driven text-to-3D generation," in Proc. Int. Conf. Comput. Vis. (ICCV), 2023.
- [16] H. Lin, J. Gao, L. Tang, T. Takikawa, X. Zeng, X. Huang, K. Kreis, S. Fidler, M.-Y. Liu, and T.-Y. Lin, "Magic3D: High-resolution text-to-3D content creation," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2023.

- [17] C. Tsalicoglou, F. Manhardt, A. Tonioni, M. Niemeyer, and F. Tombari, "TextMesh: Generation of realistic 3D meshes from text prompts," in Proc. Int. Conf. 3D Vis. (3DV), 2024.
- [18] L. Abergo, M. Morelli, and A. Guardone, "Aerodynamic shape optimization based on discrete adjoint and RBF," J. Comput. Phys., 2023.
- [19] A. Jameson, "Aerodynamic shape optimization using the adjoint method," Lectures at the Von Karman Inst., Brussels, 2003.
- [20] A. Jameson, "Efficient aerodynamic shape optimization," in Proc. AIAA/ISSMO Multidiscip. Anal. Optim. Conf., 2004.
- [21] A. Jameson, "Advances in aerodynamic shape optimization," Stanford Univ., Stanford, CA, 2004.
- [22] Van der Velden, "Aerodynamic shape optimization," AGARD-FDP-VKI Spec. Course, VKI, Rhode-Saint-Genese, Belgium, 1994.
- [23] S. Kim, S. Shankaran, and K. Leoviriyakit, "Aerodynamic shape optimization: Exploring the limits of design," in Proc. Int. Sess., GyeongJu, Korea, 2003.
- [24] A. Jameson, "Aerodynamic shape design and optimization - Status and trends," in AIAA Paper 2004-4369, 2004.
- [25] J. Reuther and A. Jameson, "Aerodynamic shape optimization of wing and wing-body configurations using control theory," NASA, 1994.
- [26] A. Lundberg, P. Hamlin, D. Shankar, A. Broniewicz, T. Walker, and C. Landström, "Automated aerodynamic vehicle shape optimization using neural networks and evolutionary optimization," SAE Int. J. Passenger Cars - Mech. Syst., 2015.
- [27] F. Muyl, L. Dumas, and V. Herbert, "Hybrid method for aerodynamic shape optimization in automotive industry," Comput. & Fluids, 2004.
- [28] A. D. Khondge, S. Sovani, and G. Verma, "Automation of vehicle aerodynamic shape exploration and optimization using integrated mesh morphing and CFD," SAE Int., 2011.
- [29] Z. Yang, Y. Jin, and Z. Gu, "Aerodynamic shape optimization method of non-smooth surfaces for aerodynamic drag reduction on a minivan," Fluids, 2021.
- [30] Y. Erdem, E. P. Gilotte, S. Harries, and C. N. Nayeri, "Aerodynamic optimization of a reduced scale model of a ground vehicle with a shape morphing technique," Fluids, 2022.
- [31] M. Xu, S. Song, X. Sun, W. Chen, and W. Zhang, "Machine learning for adjoint vector in aerodynamic shape optimization," Acta Mech. Sinica, 2021.
- [32] L. L. Wu, Y. Fu, X. B. Bu, X. R. Li, and X. L. Ma, "Research on automatic aerodynamic optimization for a SUV based on RBF model," J. Phys., 2020.
- [33] X. Han and D. W. Zingg, "An adaptive geometry parametrization for aerodynamic shape optimization," Optim. Eng., 2013.
- [34] J. Brelje and J. R. R. A. Martins, "Coupled component sizing and aerodynamic shape optimization via geometric constraints," in Proc. AIAA Aviat. Forum, 2019.
- [35] K. Wang, Z.-H. Han, K.-S. Zhang, and W.-P. Song, "An efficient geometric constraint handling method for surrogate-based aerodynamic shape optimization," Eng. Appl. Comput. Fluid Mech., 2023.
- [36] K. Bobrowski, E. Ferrer, E. Valero, and H. Barnewitz, "Aerodynamic shape optimization using geometry surrogates and adjoint method," AIAA J., 2017.
- [37] J. A. Samareh, "Aerodynamic shape optimization based on free-form deformation," in Proc. 10th AIAA/ISSMO Multidiscip. Anal. Optim. Conf., 2004.
- [38] S.-H. Yun, Y.-C. Ku, J.-H. Rho, and D.-H. Lee, "Application of function-based design method to automobile aerodynamic shape optimization," in Proc. 12th AIAA/ISSMO Multidiscip. Anal. Optim. Conf., 2008.
- [39] S. Sun, Y.-P. Chang, Q. Fu, J. Zhao, L. Ma, S. Fan, B. Li, A. Shestopalov, P. Stewart, and H. Friz, "Aerodynamic shape optimization of an SUV in early development stage using a response surface method," SAE Int. J. Passenger Cars - Mech. Syst., 2014.
- [40] Y. Zhang, W. Ding, and Y. Zhang, "Aerodynamic shape optimization based on the MIRA reference car model," SAE Tech. Paper, 2014.
- [41] X. Hu, B. Yang, Y. Lei, J. Wang, X. Li, and L. Liao, "Automotive shape optimization using the radial basis function model based on a parametric surface grid," Proc. IMechE Part D: J. Automob. Eng., 2016.
- [42] K. Song, K. H. Choo, J.-H. Kim, and D. N. Mavris, "Aerodynamic automobile shape optimization by incorporating reverse shape design method with CFD analysis," in Proc. ASME Int. Des. Eng. Tech. Conf. Comput. Inf. Eng. Conf., 2017.
- [43] K. Song, K. H. Choo, and D. Mavris, "A spline-based modeling algorithm for application to aerodynamic shape optimization based on CFD analysis," SAE Int. J. Passenger Cars - Mech. Syst., 2017.
- [44] G. Cheylan, D. Ricot, and P. Sagaut, "Shape optimization using the adjoint lattice Boltzmann method for aerodynamic applications," AIAA J., 2019.
- [45] X. Tian and J. Li, "Robust aerodynamic shape optimization using a novel multi-objective evolutionary algorithm coupled with surrogate model," Struct. Multidiscip. Optim., 2020.
- [46] Y. Zhang, J. Wu, A. X. Chang, and R. Zhang, "Image GANs meet differentiable rendering for inverse graphics and interpretable 3D neural rendering," arXiv preprint arXiv:2010.09125, 2020.